

27 Supersymmetric Gauge Theories

27.9 Gauge Theories with Extended Supersymmetry⁷

Theories with unbroken extended supersymmetry are not considered to be good candidates for realistic extensions of the standard model, because of the non-chirality of particle multiplets discussed in Section 25.4. Nevertheless gauge theories with extended supersymmetry are worth some consideration here because they have provided paradigms for the use of powerful mathematical methods to solve dynamical problems.

There are a number of special formalisms that have been proposed to construct Lagrangians with $N = 2$ extended supersymmetry [12], but fortunately we can get by with the tools already at hand. Any theory with $N = 2$ supersymmetry also has $N = 1$ supersymmetry, so its Lagrangian must be a special case of the Lagrangians already considered in this chapter. To construct a Lagrangian with $N = 2$ supersymmetry for some set of the $N = 2$ supermultiplets of particles constructed in Sections 25.4 and 25.5, we need only write down the most general Lagrangian with $N = 1$ supersymmetry whose $N = 1$ supermultiplets contain physical fields for the particles in the $N = 2$ supermultiplets, and then impose a discrete R -symmetry on the Lagrangian: a symmetry that acts differently on different components of $N = 2$ supermultiplets. The Lagrangian density will then be invariant under a second supersymmetry, whose supermultiplets are given by acting on the supermultiplets of ordinary $N = 1$ supersymmetry with the R -symmetry.

It will be convenient to choose the discrete R -transformation so that

$$Q_1 \longrightarrow Q_2, \quad Q_2 \longrightarrow -Q_1. \quad (27.9.1)$$

If the central charge were zero then the supersymmetry algebra would be invariant under an $SU(2)$ R -symmetry group, which has the transformation (27.9.1) as one finite element $\exp(i\pi\tau_2/2)$, but symmetry under the discrete symmetry is sufficient for our purposes, so we do not need to assume a zero central charge. In fact, it will turn out that the Lagrangians we construct by this method will have an $SU(2)$ R -symmetry, not just symmetry under the discrete transformation (27.9.1).

Let us first consider the renormalizable theory of the gauge bosons of a general gauge group, together with the superpartners required by $N = 2$ extended supersymmetry. We saw in Section 25.4 that in $N = 2$ global supersymmetry theories a massless gauge boson can only belong to a multiplet also containing a pair of massless fermions of each helicity $\pm 1/2$ that transform as a doublet under the $SU(2)$ R -symmetry and a pair of $SU(2)$ -singlet spinless bosons. Since $N = 2$ supersymmetry includes $N = 1$ supersymmetry, the renormalizable Lagrangian for this theory must be a special case of the general renormalizable Lagrangian density (27.4.1). One feature of this special case is that since the gauge boson belongs to the adjoint representation of the gauge group, so must also the fermions and scalar field. To furnish $N = 2$ supermultiplets of fields with the correct particle content, we must have one $N = 1$ chiral superfield Φ_A with component fields $\phi_A, \psi_A, \mathcal{F}_A$ (with ψ_A

⁷This section lies somewhat out of the book's main line of development and may be omitted in a first reading.

Majorana and ϕ_A and $\mathcal{F}_A$ both complex) for each $N = 1$ gauge multiplet V_A^μ, λ_A, D_A . We impose a discrete R -symmetry under the transformation

$$\psi_A \longrightarrow \lambda_A, \quad \lambda_A \longrightarrow -\psi_A, \quad (27.9.2)$$

(with all other fields invariant) because this is the effect of the transformation (27.9.1). Since a non-trivial superpotential would give ψ_A interactions or mass terms that are absent for λ_A , the superpotential must vanish. The Lagrangian (27.4.1) therefore takes the special form

$$\begin{aligned} \mathcal{L} = & - \sum_A (D_\mu \phi)_A^* (D^\mu \phi)_A - \frac{1}{2} \sum_A (\bar{\psi}_A (\not{D} \psi)_A) + \sum_A \mathcal{F}_A^* \mathcal{F}_A \\ & - 2\sqrt{2} \operatorname{Re} \sum_{ABC} C_{ABC} (\lambda_{AL}^T \epsilon \psi_{CL}) \phi_B^* + i \sum_{ABC} C_{ABC} \phi_B^* \phi_C D_A - \sum_A \xi_A D_A + \frac{1}{2} \sum_A D_A D_A \\ & - \frac{1}{4} \sum_A f_{A\mu\nu} f_A^{\mu\nu} - \frac{1}{2} \sum_A (\bar{\lambda}_A (\not{D} \lambda)_A) + \frac{g^2 \theta}{64\pi^2} \epsilon_{\mu\nu\rho\sigma} \sum_A f_A^{\mu\nu} f_A^{\rho\sigma}, \end{aligned} \quad (27.9.3)$$

where

$$(D_\mu \psi)_A = \partial_\mu \psi_A + \sum_{BC} C_{ABC} V_{B\mu} \psi_C, \quad (27.9.4)$$

$$(D_\mu \lambda)_A = \partial_\mu \lambda_A + \sum_{BC} C_{ABC} V_{B\mu} \lambda_C, \quad (27.9.5)$$

$$(D_\mu \phi)_A = \partial_\mu \phi_A + \sum_{BC} C_{ABC} V_{B\mu} \phi_C, \quad (27.9.6)$$

and

$$f_{A\mu\nu} = \partial_\mu V_{A\nu} - \partial_\nu V_{A\mu} + \sum_{BC} C_{ABC} V_{B\mu} V_{C\nu}. \quad (27.9.7)$$

(Recall that in the adjoint representation $(t_A)_{BC} = -iC_{ABC}$, where C_{ABC} is the real structure constant, defined as usual in this book to include factors of gauge couplings, and taken in a basis in which it is totally antisymmetric.) The Lagrangian density (27.9.3) has $N = 1$ supersymmetry with multiplets $\phi_A, \psi_A, \mathcal{F}_A$ and V_A^μ, λ_A, D_A , because it is a special case of Eq. (27.4.1), and it has an $SU(2)$ symmetry relating ψ_A and λ_A , including invariance under the finite $SU(2)$ transformation (27.9.2), so it also has a *second* independent $N = 1$ supersymmetry with multiplets $\phi_A, \lambda_A, \mathcal{F}_A$ and $V_A^\mu, -\psi_A, D_A$. It therefore satisfies the conditions imposed by $N = 2$ supersymmetry.

We can eliminate the auxiliary fields by setting them equal to the values at which the Lagrangian density (27.9.3) is stationary:

$$\mathcal{F}_A = 0, \quad D_A = -i \sum_{BC} C_{ABC} \phi_B^* \phi_C. \quad (27.9.8)$$

(We are now assuming that the Fayet–Iliopoulos constants ξ_A all vanish.) Inserting these values back in Eq. (27.9.3) gives an equivalent Lagrangian density

$$\begin{aligned}\mathcal{L} = & - \sum_A (D_\mu \phi)_A^* (D^\mu \phi)_A - \frac{1}{2} \sum_A (\bar{\psi}_A (\not{D} \psi)_A) \\ & + \sqrt{2} \sum_{ABC} C_{ABC} \left(\bar{\psi}_B \frac{1-\gamma_5}{2} \lambda_A \right) \phi_C - \sqrt{2} \sum_{ABC} C_{ABC} \left(\bar{\lambda}_A \frac{1+\gamma_5}{2} \psi_C \right) \phi_B^* - V(\phi, \phi^*) \\ & - \frac{1}{4} \sum_A f_{A\mu\nu} f_A^{\mu\nu} - \frac{1}{2} \sum_A (\bar{\lambda}_A (\not{D} \lambda)_A) + \frac{g^2 \theta}{64\pi^2} \epsilon_{\mu\nu\rho\sigma} \sum_A f_A^{\mu\nu} f_A^{\rho\sigma},\end{aligned}\tag{27.9.9}$$

where the potential is

$$V(\phi, \phi^*) = -\frac{1}{2} \sum_A \left[\sum_{BC} C_{ABC} \phi_B^* \phi_C \right]^2 = 2 \sum_A \left[\sum_{BC} C_{ABC} \operatorname{Re} \phi_B \operatorname{Im} \phi_C \right]^2. \tag{27.9.10}$$

This potential has a minimum value of zero, which is reached not only for $\phi_A = 0$ but also for any set of ϕ s for which $\sum_{BC} C_{ABC} \phi_B^* \phi_C = 0$ for all A or, in other words, for which

$$[t \cdot \operatorname{Re} \phi, t \cdot \operatorname{Im} \phi] = 0, \quad \text{where} \quad t \cdot v \equiv \sum_B t_B v_B. \tag{27.9.11}$$

That is, the minimum of the potential is reached for those scalar fields for which all generators $t \cdot \operatorname{Re} \phi$ and $t \cdot \operatorname{Im} \phi$ belong to a Cartan subalgebra of the full gauge algebra, all of whose generators commute with one another. Though all such values of ϕ give zero potential, and hence unbroken $N = 2$ supersymmetry, they are not physically equivalent, as shown for instance by the different masses they give the gauge bosons associated with the broken gauge symmetries.

One remarkable feature of extended supersymmetry is that the central charges of the supersymmetry algebra in any state can be calculated in terms of ‘charges’ in that state to which bosonic fields are coupled [13]. The easiest way to do this calculation is to use the transformation properties under ordinary $N = 1$ supersymmetry of the extended supersymmetry currents $S_r^\mu(x)$ with $r = 2, 3, \dots, N$ to calculate the anticommutators $\{Q_{1\alpha}, S_{r\beta}^\mu(x)\}$. We can then calculate the central charges from the anticommutators

$$\{Q_{1\alpha}, Q_{r\beta}\} = \int d^3x \{Q_{1\alpha}, S_{r\beta}^0(x)\}. \tag{27.9.12}$$

It turns out that the integrand on the right-hand side is a derivative with respect to space coordinates, but its integral does not vanish if the states have fields that do not vanish rapidly as $\mathbf{x} \rightarrow \infty$.

To see how this works in detail, let’s consider the case of $N = 2$ supersymmetry with an $SU(2)$ gauge symmetry and a single $N = 2$ gauge supermultiplet, with no additional matter supermultiplets. The Lagrangian here is given by Eq. (27.9.3), with A, B , and C running over the values 1, 2, 3, and

$$C_{ABC} = \epsilon \epsilon_{ABC}, \quad \xi_A = 0. \tag{27.9.13}$$

(The coupling constant here is denoted e because this is the charge with which the massless gauge field of the unbroken $U(1)$ gauge symmetry interacts.) The usual $N = 1$ supersymmetry current (distinguished now with a subscript 1) is given by Eq. (27.4.40) as

$$S_1^\mu = -\frac{1}{4} \sum_A f_{A\rho\sigma} [\gamma^\rho, \gamma^\sigma] \gamma^\mu \lambda_A - e \sum_{ABC} \epsilon_{ABC} \gamma_5 \gamma^\mu \lambda_A \phi_B^* \phi_C + \frac{1}{\sqrt{2}} \sum_A [(\not{D}\phi)_A \gamma^\mu \psi_{AR} + (\not{D}\phi^*)_A \gamma^\mu \psi_{AL}]. \quad (27.9.14)$$

We can calculate the second supersymmetry current by subjecting S_1^μ to the finite $SU(2)$ R -symmetry used above, which simply amounts to making the replacements $\psi_A \rightarrow \lambda_A$, $\lambda_A \rightarrow -\psi_A$. This gives

$$S_2^\mu = \frac{1}{4} \sum_A f_{A\rho\sigma} [\gamma^\rho, \gamma^\sigma] \gamma^\mu \psi_A + e \sum_{ABC} \epsilon_{ABC} \gamma_5 \gamma^\mu \psi_A \phi_B^* \phi_C + \frac{1}{\sqrt{2}} \sum_A [(\not{D}\phi)_A \gamma^\mu \lambda_{AR} + (\not{D}\phi^*)_A \gamma^\mu \lambda_{AL}]. \quad (27.9.15)$$

It will be enough for our purposes (and somewhat easier) to calculate only the change in the right-handed part of this current under an $N = 1$ supersymmetry transformation. After setting the auxiliary fields equal to their equilibrium values

$$\mathcal{F}_A = 0, \quad D_A = -ie \sum_{BC} \epsilon_{ABC} \phi_B^* \phi_C,$$

we find

$$\begin{aligned} \delta S_{2R}^\mu &= \frac{\sqrt{2}}{4} \sum_A f_{A\rho\sigma} [\gamma^\rho, \gamma^\sigma] \gamma^\mu (\not{D}\phi)_A \alpha_R \\ &\quad - \sqrt{2}e \sum_{ABC} \epsilon_{ABC} \gamma^\mu (\not{D}\phi)_A \alpha_R \phi_B^* \phi_C - \frac{\sqrt{2}}{4} f_{A\rho\sigma} (\not{D}\phi)_A \gamma^\mu [\gamma^\rho, \gamma^\sigma] \alpha_R \\ &\quad - \sqrt{2}e \sum_{ABC} \epsilon_{ABC} \phi_B^* \phi_C (\not{D}\phi)_A \gamma^\mu \alpha_R + \dots, \end{aligned}$$

where the dots denote terms bilinear in fermion fields, which do not concern us here because we are interested in effects of long range boson fields. We can combine terms by using the Dirac anticommutation relations and the identity

$$[\gamma^\rho, \gamma^\sigma][\gamma^\mu, \gamma^\nu] + [\gamma^\mu, \gamma^\nu][\gamma^\rho, \gamma^\sigma] = -8\eta^{\mu\rho}\eta^{\nu\sigma} + 8\eta^{\sigma\mu}\eta^{\rho\nu} + 8i\epsilon^{\mu\nu\rho\sigma}\gamma_5$$

and find

$$\begin{aligned} \delta S_{2R}^\mu &= -2\sqrt{2} \sum_A f_A^{\mu\nu} (D_\nu \phi)_A \alpha_R - i\sqrt{2} \sum_A \epsilon^{\mu\nu\rho\sigma} f_{A\rho\sigma} (D_\nu \phi)_A \alpha_R \\ &\quad - 2\sqrt{2}e \sum_{ABC} \epsilon_{ABC} \phi_B^* \phi_C (D^\mu \phi)_A \alpha_R + \dots. \end{aligned}$$

In order to write this as a derivative, we need to use the Yang–Mills field equations given by Eqs. (15.3.6), (15.3.7), and (15.3.9):

$$D_\nu f_A^{\mu\nu} = J_A^\mu = e \sum_{BC} \epsilon_{ABC} ((D^\mu \phi)_B^* \phi_C - \phi_B^* (D^\mu \phi)_C),$$

$$\epsilon_{\mu\nu\rho\sigma} (D^\nu f^{\rho\sigma})_A = 0.$$

These allow us to write δS_{2R}^μ as a total derivative

$$\delta S_{2R}^\mu = D_\nu X^{\mu\nu} \alpha_R, \quad (27.9.16)$$

where

$$X^{\mu\nu} = -2\sqrt{2} \sum_A f_A^{\mu\nu} \phi_A - i\sqrt{2} \sum_A \epsilon^{\mu\nu\rho\sigma} f_{A\rho\sigma} \phi_A + \dots, \quad (27.9.17)$$

with the dots again indicating irrelevant terms involving fermion fields. Eq. (26.1.18) allows us to write Eq. (27.9.16) as an anticommutation relation

$$\{Q_{R\alpha}, S_{R\beta}^\mu\} = i \left[\epsilon \left(\frac{1 - \gamma_5}{2} \right) \right]_{\alpha\beta} D_\nu X^{\mu\nu}. \quad (27.9.18)$$

Since $X^{\mu\nu}$ is a gauge-invariant quantity, its gauge-covariant derivative is the same as its ordinary derivative. Also, $X^{\mu\nu}$ is antisymmetric, so $D_\nu X^{0\nu} = \partial_i X^{0i}$. From Eqs. (27.9.12) and (27.9.18), we have at last

$$\{Q_{R\alpha}, Q_{R\beta}\} = i \left[\epsilon \left(\frac{1 - \gamma_5}{2} \right) \right]_{\alpha\beta} \int dS_i X^{0i}, \quad (27.9.19)$$

with the integral taken over a large closed surface enclosing the system in question, with surface area differential dS taken normal to the surface. Comparing this with Eq. (25.2.38) gives the central charge

$$Z_{12} = -i \int dS_i X^{0i}. \quad (27.9.20)$$

If we choose a gauge in which ϕ_A (almost everywhere) has only the constant non-vanishing component $\phi_3 = v$, then

$$\sum_A f_A^{0i} \phi_A = -v E^i, \quad \frac{1}{2} \sum_A \epsilon^{0i\rho\sigma} f_{A\rho\sigma} \phi_A = v B^i, \quad (27.9.21)$$

where $\mathbf{E}$ and $\mathbf{B}$ are the electric and magnetic fields associated with the unbroken $U(1)$ subgroup of the $SU(2)$ gauge group. Therefore the central charge (27.9.20) here is

$$Z_{12} = 2\sqrt{2}v[iq - \mathcal{M}], \quad (27.9.22)$$

where q and $\mathcal{M}$ are the electric charge and magnetic monopole moment, defined by

$$q = \int dS_i E^i, \quad \mathcal{M} = \int dS_i B^i. \quad (27.9.23)$$

As discussed in Section 23.3, this theory, with $SU(2)$ gauge symmetry spontaneously broken by the expectation value of an $SU(2)$ triplet of scalars, is one in which magnetic monopoles actually do occur.

The application of the results of Section 27.4 to the Lagrangian density (27.9.3) shows that, after the spontaneous breaking of the $SU(2)$ gauge symmetry, this theory will contain elementary particles of charge $\pm e$, zero magnetic monopole moment, and tree-approximation mass $M = \sqrt{2}|ev|$. Specifically, for each sign of the charge there is one such particle of spin 1, two of them with spin 1/2, and one with spin 0. It is a striking consequence of the results obtained here that *the mass value $\sqrt{2}|ev|$ is exact, being unaffected by either radiative corrections or non-perturbative effects, provided that the quantity v is defined by Eq. (27.9.22) for the central charge [13].*

To see this, note that the massive one-particle states for each sign of the charge are ‘short’ $N = 2$ supermultiplets, which, as shown at the end of Section 25.5, have masses that saturate the lower bound (25.5.24):

$$M = |Z_{12}|/2. \quad (27.9.24)$$

Even if we did not trust the tree approximation to give us the precise value of the particle masses, we would not expect corrections to this approximation to turn short multiplets into the full multiplets, with many more states, that could have larger masses, so we can be confident that Eq. (27.9.24) is exactly valid. For particles with electric charge $q = \pm e$ and zero magnetic monopole moment, Eq. (27.9.20) gives $Z_{12} = \pm 2\sqrt{2}ive$, so Eq. (27.9.24) tells us that their masses are

$$M = \sqrt{2}|ev|. \quad (27.9.25)$$

This is the result found in the tree approximation, but now we see that it is exact.

The semi-classical calculations described in Section 23.3 show that electrically neutral magnetic monopoles in this theory have monopole strengths⁸

$$\mathcal{M} = \frac{4\pi\nu}{e}, \quad (27.9.26)$$

with ν the winding number, a positive or negative integer. The formula (27.9.22) for the central charge together with the inequality (25.5.24) thus give a lower bound on the monopole masses

$$M \geq \frac{4\pi\sqrt{2}|\nu v|}{|e|}. \quad (27.9.27)$$

Interestingly, this is the same as the Bogomol’nyi lower bound [14] on monopole energies derived in Section 23.3.⁹ In fact, the monopole solution for $\nu = 1$ that was described in Section 23.3 saturates this bound. More generally, the ‘dyons’ of this theory [15], particles with both charge and magnetic moment, have masses given by [16]

$$M = 2|v|\sqrt{q^2 + \mathcal{M}^2}, \quad (27.9.28)$$

⁸Note that the magnetic moment $\mathcal{M}$ defined by Eq. (27.9.23) is related to the magnetic moment g defined in Section 23.3 by $\mathcal{M} = 4\pi g$.

⁹The canonically normalized field with a non-vanishing vacuum expectation value is $\sqrt{2}\text{Re } \phi_3$ (for real v), so the quantity $\langle \phi \rangle$ appearing in the Bogomol’nyi inequality (23.3.19) is $\sqrt{2}v$.

which again is the minimum value allowed by Eqs. (25.5.24) and (27.9.20). Indeed, all of the known particles in this theory have masses given in the semi-classical limit by Eq. (27.9.28) [17].

Returning now to general $N = 2$ gauge theories, we may also include additional ‘matter’ fields in the Lagrangian. For simplicity, we will restrict ourselves to ‘short’ massive hypermultiplets (with central charge $\mathcal{Z}$ saturating the inequality (25.5.24)), each consisting of a single fermion of spin 1/2 and an $SU(2)$ doublet of spin 0 particles, together with distinct antiparticles. This is the same spin content as is given under $N = 1$ supersymmetry by pairs of left-chiral scalar superfields Φ'_n and Φ''_n , together with their right-chiral adjoints, with the complex scalar field components ϕ'_n and ϕ''_n and their adjoints forming pairs of $SU(2)$ doublets, and the spinor fields all $SU(2)$ singlets. (We are using primes and double primes to distinguish these superfields and their components from Φ_A and its components.) If some of these hypermultiplets Φ'_n and Φ''_n are non-neutral under the gauge group, then a superpotential is allowed, of the form:¹⁰

$$f(\Phi, \Phi', \Phi'') = \frac{1}{2} \sum_{Anm} (s_A)_{nm} \Phi'_n \Phi''_m \Phi_A + \frac{1}{2} \sum_{nm} \mu_{nm} \Phi'_n \Phi''_m. \quad (27.9.29)$$

To the Lagrangian density (27.9.3) we then must add a Lagrangian density for these hypermultiplets, given by the first eight terms on the right-hand side of Eq. (27.4.1), and find

¹⁰There still cannot be any terms in the superpotential that are of second or higher order in the Φ_A for the same reason as before: such terms would lead to scalar couplings or masses for the ψ_A , with no corresponding couplings or masses for their $SU(2)$ partners λ_A . Also, there cannot be any term that is trilinear in the Φ'_n and/or Φ''_n , because then the term in Eq. (27.4.1) involving the product of fermion bilinears with second derivatives of the superpotential would lead to a coupling of the $SU(2)$ -singlet fermions to $SU(2)$ doublet fields ϕ'_n or ϕ''_n . Thus the only trilinear interaction terms in the superpotential must involve one factor of Φ_A and two factors of Φ'_n and/or Φ''_n . There cannot be any trilinear terms involving a Φ_A and two Φ'_n s or two Φ''_n s, because that would give the $SU(2)$ singlet auxiliary fields $\mathcal{F}_A$ an interaction with $SU(2)$ -triplet products $\phi'_n \phi'_m$ or $\phi''_n \phi''_m$, and there cannot be any bilinear terms involving two Φ'_n s or two Φ''_n s, because that would yield $SU(2)$ -triplet mass terms $(\psi'^T_{nL} \epsilon \psi'_{mL})$ or $(\psi''^T_{nL} \epsilon \psi''_{mL})$. The only remaining allowed bilinear or trilinear terms are of the form (27.9.29).

a total Lagrangian density

$$\begin{aligned}
\mathcal{L} = & - \sum_n (D_\mu \phi')_n^* (D^\mu \phi')_n - \sum_n (D_\mu \phi'')_n^* (D^\mu \phi'')_n - \sum_A (D_\mu \phi)_A^* (D^\mu \phi)_A \\
& - \frac{1}{2} \sum_n (\bar{\psi}'_n (\not{D} \psi')_n) - \frac{1}{2} \sum_n (\bar{\psi}''_n (\not{D} \psi'')_n) - \frac{1}{2} \sum_A (\bar{\psi}_A (\not{D} \psi)_A) - \frac{1}{2} \sum_A (\bar{\lambda}_A (\not{D} \lambda)_A) \\
& + \sum_n \mathcal{F}_n'^* \mathcal{F}_n' + \sum_n \mathcal{F}_n''^* \mathcal{F}_n'' + \sum_A \mathcal{F}_A^* \mathcal{F}_A \\
& - \text{Re} \sum_{Anm} (s_A)_{nm} \phi_A (\psi_{nL}'^T \epsilon \psi_{mL}'') - 2\sqrt{2} \text{Re} \sum_{ABC} C_{ABC} (\lambda_{AL}'^T \epsilon \psi_{CL}) \phi_B^* \\
& - \text{Re} \sum_{Anm} (s_A)_{nm} \phi_n' (\psi_{mL}''^T \epsilon \psi_{AL}) - \text{Re} \sum_{Anm} (s_A)_{nm} \phi_m'' (\psi_{nL}'^T \epsilon \psi_{AL}) \\
& + 2\sqrt{2} \text{Im} \sum_{Anm} (t_A')_{mn} (\psi_{nL}'^T \epsilon \lambda_{AL}) \phi_m'^* + 2\sqrt{2} \text{Im} \sum_{Anm} (t_A'')_{mn} (\psi_{nL}''^T \epsilon \lambda_{AL}) \phi_m''^* \\
& + \text{Re} \sum_{Anm} (s_A)_{nm} \phi_A \phi_n' \mathcal{F}_m'' + \text{Re} \sum_{Anm} (s_A)_{nm} \phi_A \phi_m'' \mathcal{F}_n' + \text{Re} \sum_{Anm} (s_A)_{nm} \phi_n' \phi_m'' \mathcal{F}_A \\
& + \text{Re} \sum_{nm} \mu_{nm} \phi_n' \mathcal{F}_m'' + \text{Re} \sum_{nm} \mu_{nm} \phi_m'' \mathcal{F}_n' - \text{Re} \sum_{nm} \mu_{nm} (\psi_{nL}'^T \epsilon \psi_{mL}'') \\
& - \sum_{Anm} (t_A')_{nm} \phi_n'^* \phi_m' D_A - \sum_{Anm} (t_A'')_{nm} \phi_n''^* \phi_m'' D_A + i \sum_{ABC} C_{ABC} \phi_B^* \phi_C D_A \\
& - \sum_A \xi_A D_A + \frac{1}{2} \sum_A D_A D_A - \frac{1}{4} \sum_A f_{A\mu\nu} f_A^{\mu\nu} + \frac{g^2 \theta}{64\pi^2} \epsilon_{\mu\nu\rho\sigma} \sum_A f_A^{\mu\nu} f_A^{\rho\sigma},
\end{aligned} \tag{27.9.30}$$

where the $(t_A')_{nm}$ and $(t_A'')_{nm}$ are the matrices (including coupling constant factors) representing the gauge group on the left-chiral scalar superfields Φ_n' and Φ_n'' , respectively. The Yukawa couplings between fermions and scalars have a discrete R -symmetry under the transformation

$$\lambda_{AL} \longrightarrow -\psi_{AL}, \quad \psi_{AL} \longrightarrow +\lambda_{AL}, \quad \phi_n'' \longrightarrow -\phi_n'^*, \quad \phi_n' \longrightarrow \phi_n''^*, \tag{27.9.31}$$

provided that

$$s_A = -2\sqrt{2}i t_A'^T = +2\sqrt{2}i t_A''^T. \tag{27.9.32}$$

(Note in particular that Eq. (27.9.32) requires the representations of the gauge group furnished by Φ_n' and Φ_n'' to be complex conjugates.) This is also a symmetry of all the other terms in the Lagrangian density (27.9.30), except for those involving the auxiliary fields.

It is not possible to extend the symmetry under the transformation (27.9.31) to the auxiliary fields, but the symmetry appears after the auxiliary fields are eliminated.¹¹ After setting D_A , $\mathcal{F}_n'$, and $\mathcal{F}_n''$ equal to values at which the Lagrangian density is stationary, and

¹¹After elimination of the auxiliary fields, the resulting action is invariant under the original $N = 2$ supersymmetry transformation only ‘on-shell’—that is, only up to terms that vanish when the fields satisfy the interacting field equations. This doesn’t hurt, because there still are two conserved supersymmetry currents whose integrated time-components satisfy the $N = 2$ supersymmetry anticommutation relations when the fields of which they are composed are required to satisfy the field equations of the Heisenberg picture. ‘Off-shell’ formulations of $N = 2$ supersymmetry exist, but are subject to various complications [18].

combining D - and $\mathcal{F}$ -terms, the Lagrangian density (with s_A and t_A'' given by Eq. (27.9.32), and ξ_A taken to vanish) takes the form

$$\begin{aligned}
\mathcal{L} = & - \sum_n (D_\mu \phi')_n^* (D^\mu \phi')_n - \sum_n (D_\mu \phi'')_n^* (D^\mu \phi'')_n - \sum_A (D_\mu \phi)_A^* (D^\mu \phi)_A \\
& - \frac{1}{2} \sum_n (\bar{\psi}'_n (\not{D} \psi')_n) - \frac{1}{2} \sum_n (\bar{\psi}''_n (\not{D} \psi'')_n) - \frac{1}{2} \sum_A (\bar{\psi}_A (\not{D} \psi)_A) - \frac{1}{2} \sum_A (\bar{\lambda}_A (\not{D} \lambda)_A) \\
& - 2\sqrt{2} \text{Im} \sum_{Anm} (t'_A)_{mn} \phi_A (\psi_{nL}'^T \epsilon \psi_{mL}'') - 2\sqrt{2} \text{Re} \sum_{ABC} C_{ABC} (\lambda_{AL}^T \epsilon \psi_{CL}) \phi_B^* \\
& - 2\sqrt{2} \text{Im} \sum_{Anm} (t'_A)_{mn} \phi_n' (\psi_{mL}''^T \epsilon \psi_{AL}) - 2\sqrt{2} \text{Im} \sum_{Anm} (t'_A)_{mn} \phi_m'' (\psi_{nL}'^T \epsilon \psi_{AL}) \\
& + 2\sqrt{2} \text{Im} \sum_{Anm} (t'_A)_{mn} (\psi_{nL}'^T \epsilon \lambda_{AL}) \phi_m'^* - 2\sqrt{2} \text{Im} \sum_{Anm} (t'_A)_{nm} (\psi_{nL}''^T \epsilon \lambda_{AL}) \phi_m''^* \\
& - \frac{1}{4} \sum_A f_{A\mu\nu} f_A^{\mu\nu} + \frac{g^2 \theta}{64\pi^2} \epsilon_{\mu\nu\rho\sigma} \sum_A f_A^{\mu\nu} f_A^{\rho\sigma} \\
& - \sum_{ABnm} \{t'_A, t'_B\}_{mn} \phi_A \phi_B^* (\phi_n'^* \phi_m' + \phi_n''^* \phi_m'') \\
& - \frac{1}{2} \sum_A \left[\sum_{nm} (t'_A)_{nm} (\phi_n'^* \phi_m' - \phi_n''^* \phi_m'') \right]^2 \\
& + \frac{1}{2} \sum_{ABCDE} C_{ABC} C_{ADE} \phi_B^* \phi_C \phi_D^* \phi_E - 2 \sum_A \left| \sum_{nm} (t'_A)_{nm} \phi_n' \phi_m'' \right|^2 \\
& - 4 \text{Re} \sum_{nm} (t'_A \mu)_{nm} \phi_n'^* \phi_m' - 4 \text{Re} \sum_{nm} (\mu t'_A)_{nm} \phi_n'' \phi_m''^* \\
& - 2 \sum_{nm} (\mu^\dagger \mu)_{nm} \phi_n'^* \phi_m' - 2 \sum_{nm} (\mu \mu^\dagger)_{nm} \phi_n'' \phi_m''^* .
\end{aligned} \tag{27.9.33}$$

The last five lines on the right-hand side come from the terms in Eq. (27.9.30) involving auxiliary fields, and now these too are invariant under the discrete transformation (27.9.31), provided that

$$[t'_A, \mu] = [\mu^\dagger, \mu] = 0. \tag{27.9.34}$$

We can now go a step further and consider the case of $N = 4$ extended global supersymmetry. (As remarked in Section 25.4, $N = 3$ supersymmetry is the same as $N = 4$ supersymmetry.) The only massless multiplets of $N = 4$ supersymmetry that do not contain gravitons or gravitinos consist of a single particle of helicity 1, an $SU(4)$ quartet of particles of helicity 1/2, and an $SU(4)$ sextet of particles of helicity 0, together with their CPT-conjugates of opposite helicity. There is one such supermultiplet for each generator t_A of the gauge group. These particles can be grouped into supermultiplets of $N = 2$ supersymmetry: for each t_A there is one gauge supermultiplet consisting of one particle of helicity 1, two particles of helicity $\pm 1/2$, and one particle of helicity 0, together with their CPT-conjugates of opposite helicity, plus two hypermultiplets, each consisting of one particle of each helicity $\pm 1/2$ and two particles of helicity 0. The $N = 2$ gauge superfield

consists of an $N = 1$ gauge superfield V_A and a left-chiral scalar superfield Φ_A and its complex conjugate, while the two $N = 2$ hypermultiplets consist of two additional left-chiral scalar superfields Φ'_A and Φ''_A and their complex conjugates.

Since $N = 4$ supersymmetry includes $N = 2$ supersymmetry, the Lagrangian density after elimination of the auxiliary fields of $N = 1$ supersymmetry [18] must be a special case of Eq. (27.9.33), but with the labels n, m , etc. running over the indices A, B, C , etc. of the adjoint representation. Also, the coefficient μ_{nm} in the superpotential (27.9.29) must vanish here, because otherwise Eq. (27.9.33) would contain terms quadratic in the fermion fields ψ'_A and ψ''_A , with no counterpart for their $N = 4$ superpartners λ_A and ψ_A . Also setting $(t'_A)_{BC}$ equal to the generators $-iC_{ABC}$ in the adjoint representation, we find that the Lagrangian density must take the form

$$\begin{aligned}
\mathcal{L} = & - \sum_A (D_\mu \phi')_A^* (D^\mu \phi')_A - \sum_A (D_\mu \phi'')_A^* (D^\mu \phi'')_A - \sum_A (D_\mu \phi)_A^* (D^\mu \phi)_A \\
& - \frac{1}{2} \sum_A (\bar{\psi}'_A (\not{D} \psi')_A) - \frac{1}{2} \sum_A (\bar{\psi}''_A (\not{D} \psi'')_A) - \frac{1}{2} \sum_A (\bar{\psi}_A (\not{D} \psi)_A) - \frac{1}{2} \sum_A (\bar{\lambda}_A (\not{D} \lambda)_A) \\
& - 2\sqrt{2} \operatorname{Re} \sum_{ABC} C_{ABC} \phi_A (\psi'^T_{BL} \epsilon \psi''_{CL}) - 2\sqrt{2} \operatorname{Re} \sum_{ABC} C_{ABC} (\lambda^T_{AL} \epsilon \psi_{CL}) \phi_B^* \\
& - 2\sqrt{2} \operatorname{Re} \sum_{ABC} C_{ABC} \phi'_B (\psi''^T_{CL} \epsilon \psi_{AL}) - 2\sqrt{2} \operatorname{Re} \sum_{ABC} C_{ABC} \phi''_C (\psi'^T_{BL} \epsilon \psi_{AL}) \\
& + 2\sqrt{2} \operatorname{Re} \sum_{ABC} C_{ABC} (\psi'^T_{BL} \epsilon \lambda_{AL}) \phi_C^* + 2\sqrt{2} \operatorname{Re} \sum_{ABC} C_{ABC} (\psi''^T_{BL} \epsilon \lambda_{AL}) \phi_C^* \\
& - \frac{1}{4} \sum_A f_{A\mu\nu} f_A^{\mu\nu} + \frac{g^2 \theta}{64\pi^2} \epsilon_{\mu\nu\rho\sigma} \sum_A f_A^{\mu\nu} f_A^{\rho\sigma} - V,
\end{aligned} \tag{27.9.35}$$

where the potential is

$$\begin{aligned}
V = & \sum_{ABCDE} C_{ADE} C_{BCE} (\phi_A \phi_B^* + \phi_B \phi_A^*) (\phi'_C \phi_D^* + \phi''^* \phi''_D) \\
& + \frac{1}{2} \sum_A \left| \sum_{BC} C_{ABC} (\phi_B^* \phi'_C - \phi_B'' \phi''^*_C) \right|^2 \\
& - \frac{1}{2} \sum_{ABCDE} C_{ABC} C_{ADE} \phi_B^* \phi_C \phi_D^* \phi_E + 2 \sum_A \left| \sum_{BC} C_{ABC} \phi'_B \phi''_C \right|^2.
\end{aligned} \tag{27.9.36}$$

With no further constraints needed, this Lagrangian has an $SU(4)$ R -symmetry, which implies that it is invariant under $N = 4$ supersymmetry. To see this, we need to use the Jacobi identity to write the cross-term in the second line of the right-hand side of Eq. (27.9.36) in the form

$$\begin{aligned}
\sum_{ABCDE} C_{ABC} C_{ADE} \phi_B^* \phi'_C \phi_D''^* \phi_E'' &= - \sum_{ABCDE} C_{ABC} C_{ADE} \phi_B^* \phi'_D \phi_E''^* \phi_C'' \\
&- \sum_{ABCDE} C_{ABC} C_{ADE} \phi_B^* \phi'_E \phi_C''^* \phi_D'',
\end{aligned}$$

which allows us to write the potential (27.9.36) in a form symmetric among the scalars and their adjoints

$$\begin{aligned}
V = & \sum_A \left| \sum_{BC} C_{ABC} \phi_B^* \phi_C' \right|^2 + \sum_A \left| \sum_{BC} C_{ABC} \phi_B^* \phi_C'' \right|^2 + \sum_A \left| \sum_{BC} C_{ABC} \phi_B \phi_C' \right|^2 \\
& + \sum_A \left| \sum_{BC} C_{ABC} \phi_B \phi_C'' \right|^2 + \sum_A \left| \sum_{BC} C_{ABC} \phi_B'^* \phi_C'' \right|^2 \\
& + \sum_A \left| \sum_{BC} C_{ABC} \phi_B' \phi_C'' \right|^2 + \frac{1}{2} \sum_A \left| \sum_{BC} C_{ABC} \phi_B' \phi_C'^* \right|^2 \\
& + \frac{1}{2} \sum_A \left| \sum_{BC} C_{ABC} \phi_B'' \phi_C'^* \right|^2 + \frac{1}{2} \sum_A \left| \sum_{BC} C_{ABC} \phi_B \phi_C^* \right|^2.
\end{aligned} \tag{27.9.37}$$

Now to make the $SU(4)$ symmetry apparent, we introduce an $SU(4)$ notation for the fields. We assemble the left-handed fermion fields into an $SU(4)$ vector:

$$\psi_{1AL} \equiv \psi_{AL}, \quad \psi_{2AL} \equiv \lambda_{AL}, \quad \psi_{3AL} \equiv \psi'_{AL}, \quad \psi_{4AL} \equiv \psi''_{AL}. \tag{27.9.38}$$

In order for the fermion kinematic terms in the Lagrangian density to be $SU(4)$ -invariant, we then must assemble the right-handed fermion fields into a contragredient vector:

$$\psi_{AR}^1 \equiv \psi_{AR}, \quad \psi_{AR}^2 \equiv \lambda_{AR}, \quad \psi_{AR}^3 \equiv \psi'_{AR}, \quad \psi_{AR}^4 \equiv \psi''_{AR}. \tag{27.9.39}$$

The Majorana condition on the fermion fields then takes the $SU(4)$ -invariant form

$$(\psi_{iAL})^* = -\beta \epsilon \psi_{AR}^i, \tag{27.9.40}$$

with indices i, j , etc. running over the values 1, 2, 3, 4. In order for the Yukawa couplings between the fermion and scalar fields to be $SU(4)$ -invariant, we must give the scalars the transformation properties of an antisymmetric $SU(4)$ tensor

$$\begin{aligned}
\phi_A^{12} &\equiv \phi_A^*, & \phi_A^{13} &\equiv \phi_A'', & \phi_A^{14} &\equiv -\phi_A', \\
\phi_A^{23} &\equiv -\phi_A'^*, & \phi_A^{24} &\equiv -\phi_A'', & \phi_A^{34} &\equiv \phi_A,
\end{aligned} \tag{27.9.41}$$

which also obeys an $SU(4)$ -invariant reality condition

$$(\phi_A^{ij})^* = \frac{1}{2} \sum_{kl} \epsilon_{ijkl} \phi_A^{kl}. \tag{27.9.42}$$

The whole Lagrangian density (27.9.35) can then be written in the manifestly $SU(4)$ -invariant form

$$\begin{aligned}
\mathcal{L} = & -\frac{1}{2} \sum_{Aij} (D_\mu \phi^{ij})_A (D^\mu \phi^{ij})_A^* \\
& -\frac{1}{2} \sum_{Ai} (\psi_{iAL}^T \epsilon (\not{D} \psi_R^i)_A) + \frac{1}{2} \sum_{Ai} (\psi_{AR}^{iT} \epsilon (\not{D} \psi_{iL})_A) \\
& -\sqrt{2} \operatorname{Re} \sum_{ABCij} C_{ABC} \phi_A^{ij} (\psi_{iBL}^T \epsilon \psi_{jCL}) - V \\
& -\frac{1}{4} \sum_A f_{A\mu\nu} f_A^{\mu\nu} + \frac{g^2 \theta}{64\pi^2} \epsilon_{\mu\nu\rho\sigma} \sum_A f_A^{\mu\nu} f_A^{\rho\sigma},
\end{aligned} \tag{27.9.43}$$

where the potential is

$$V = \frac{1}{8} \sum_{Aijkl} \left| \sum_{BC} C_{ABC} \phi_B^{ij} \phi_C^{kl} \right|^2. \quad (27.9.44)$$

The potential has a minimum value zero, so that supersymmetry is not broken in this theory. The minimum is reached when the generators $\sum_A t_A \phi_A^{ij}$ all commute with one another.

For zero theta angle, the gauge theories with a simple gauge group and either $N = 2$ or $N = 4$ supersymmetry have just a single coupling constant, the gauge coupling constant g . Since these theories have $N = 1$ supersymmetry, they share the property discussed in Section 27.6, that the only infinity in higher orders of perturbation theory is in a one-loop correction to this coupling.¹² The function $\beta(g)$ in the renormalization group equation $\mu dg/d\mu = \beta(g)$ is then given to all orders of perturbation theory by the one-loop formula (18.7.2), with a suitable correction for the presence of scalar fields:

$$\beta(g) = -\frac{g^3}{4\pi^2} \left(\frac{11}{12} C_1 - \frac{1}{6} C_2^f - \frac{1}{12} C_2^s \right), \quad (27.9.45)$$

where

$$\begin{aligned} \sum_{AB} C_{ABC} C_{ABD} &= g^2 C_1 \delta_{CD}, \\ [\text{Tr}(t_C t_D)]_{\text{Majorana fermions}} &= g^2 C_2^f \delta_{CD}, \\ [\text{Tr}(t_C t_D)]_{\text{complex scalars}} &= g^2 C_2^s \delta_{CD}. \end{aligned} \quad (27.9.46)$$

In general theories with $N = 2$ supersymmetry we have two Majorana fermions λ_A and ψ_A in the adjoint representation and H pairs of Majorana fermions ψ'_n and ψ''_n whose left- and right-handed parts are in representations with generators either t'_A or $-t_A^T$, so

$$C_2^f = 2C_1 + 2HC'_2, \quad (27.9.47)$$

where C'_2 is defined by

$$\text{Tr } t'_C t'_D = g^2 C'_2 \delta_{CD}. \quad (27.9.48)$$

Also, we have one complex scalar ϕ_A in the adjoint representation and H pairs of complex scalars ϕ'_n and ϕ''_n in representations with generators t'_A or $-t_A^T$, so

$$C_2^s = C_1 + 2HC'_2. \quad (27.9.49)$$

The beta function (27.9.45) is therefore

$$\beta(g) = -\frac{g^2}{8\pi^2} (C_1 - HC'_2). \quad (27.9.50)$$

The case of $N = 4$ supersymmetry is just the special case with $H = 1$ pairs of $N = 2$ hypermultiplets in the adjoint representation, with $C'_2 = C_1$, so in this case the beta function vanishes. This is therefore a finite theory, with no renormalizations at all [19].

¹²The trilinear term in the superpotential (27.9.29) is proportional to the gauge coupling, and is therefore renormalized, despite the no-renormalization theorem of Section 27.6. This is because here we are renormalizing the left-chiral scalar superfields Φ_A, Φ'_n , and Φ''_n as well as the gauge superfield V_A so as to keep them canonically normalized. The bilinear term in Eq. (27.9.29) is renormalized for the same reason.

Gauge theories with $N = 4$ supersymmetry have another remarkable property, known as *duality*. This was first conjectured by Montonen and Olive [17] for purely bosonic theories in which a simple gauge group is spontaneously broken to a $U(1)$ electromagnetic gauge group. They noticed that semi-classical calculations (of the sort described in Section 23.3) give the mass of particles with charge $q = ne$ and magnetic monopole moment $\mathcal{M} = 4\pi m/e$ (with n and m integers of any sign) as

$$M = \sqrt{2} \left| v \left(ne + \frac{4\pi im}{e} \right) \right|, \quad (27.9.51)$$

which is invariant under the transformations

$$m \longrightarrow n, \quad n \longrightarrow -m, \quad e \longleftrightarrow 4\pi/e. \quad (27.9.52)$$

On this basis, they suggested that the theory with a weak gauge coupling e is fully equivalent to one with a strong gauge coupling $4\pi/e$. Neither the purely bosonic theory nor the simplest versions of the $N = 1$ and $N = 2$ extended supersymmetry theories really have this property; [20] for one thing, the massive charged elementary vector bosons of the broken gauge symmetries have spin 1, while all of the monopoles and dyons have spins $1/2$ or 0 . (We will see in Section 29.5 that $N = 2$ theories do have a duality property of a more subtle sort.) But for $N = 4$ supersymmetry the monopole states form multiplets with one particle of spin 1, four particles of spin $1/2$, and two particles of spin 0 , just like the elementary particles [20]. Evidence has accumulated [21] that $N = 4$ supersymmetric gauge theories are indeed invariant under the interchange of electric and magnetic quantum numbers and of e with $4\pi/e$. The equivalence of theories with large and small coupling constants has become an increasingly important theme in string theory, but this is beyond the scope of this book.